Customer Stories / Software & Internet / United States

2024

DoorDash Processed More Than 10 Million Sweepstakes Entri

Learn how DoorDash rapidly scaled the implementation of an event-driven architecture using AWS Lambda and Amazon API Gateway.

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>10 million

entries processed successfully

725,000 requests

to Amazon API Gateway per minute received

8 million requests

to Amazon CloudFront per minute received

Overview

DoorDash wanted to highlight its ability to deliver nearly anything while standing out among the dozens of high-profile TV advertisements that would show during the 2023 American football championship. So, the company created a 30-second interactive ad that prompted viewers to enter a promotional code online for a chance to win items related to each and every national commercial that aired. DoorDash expected a huge number of viewers to enter the sweepstakes, so it used Amazon Web Services (AWS) to power a dedicated website that would be scalable, reliable, and fast. The company also worked with Wildlife, a creative agency, to build an event-driven architecture (EDA) that would scale rapidly to handle millions of entries while remaining cost efficient. Using serverless services on AWS, DoorDash processed more than 10 million entries in five hours with virtually no downtime and no widespread issues. The ad received multiple Cannes Lions awards, including a Titanium Grand Prix in addition to the Cannes Gold Lion and Silver Lion for Direct Engagement.

Opportunity | Using AWS Serverless Services to Build a Scalable, Reliable Website for DoorDash

DoorDash is a local commerce platform that operates in more than 30 countries. To capture the attention of viewers, DoorDash launched a sweepstakes for one grand prize containing dozens of products seen in football championship TV ads, prompting viewers to enter a promotional code from its commercial on a microsite. With more than 100 million viewers, scalability and reliability were essential.

DoorDash powers most of its infrastructure using AWS because of its scalability and variety of services, and Wildlife has been building solutions on AWS since 2014. So, the companies chose to use AWS to build a scalable microsite for the sweepstakes. “We’ve invested heavily in using AWS services for nearly all our projects,” says Pavel Zagoskin, Technical Director at Wildlife.

For DoorDash’s sweepstakes website, Wildlife designed a multi–AWS Region EDA—an architecture pattern that is built from small, decoupled services that operate according to near real-time changes in state. “We chose a serverless approach because we didn’t want to deal with any servers or hardware management,” says Zagoskin. With a serverless architecture, the website would be able to handle millions of concurrent requests with a pay-as-you-go model.

DoorDash received support from [AWS Countdown Premium](), a service for optimizing business-critical events, launches, and migrations on AWS, 11 days before finishing the website. “It was invaluable to h support engineers from AWS,” says Brian Alexander, Technical Program Manager at DoorDash would get back to us very quickly with resolutions or suggestions.” The DoorDash and Wildlife teams worked closely alongside AWS throughout the project. “We saw the power and potential of collaborating with the experts at AWS and going beyond our knowledge for new or unfamiliar challenges,” says Jake Friedman, Cofounder and President of Wildlife.

Solution | Processing More Than 10 Million API Requests on AWS

Wildlife based the design on AWS Lambda, which businesses can use to run code without operating servers or clusters. To manage its API, DoorDash uses Amazon API Gateway, which makes it simple to create, maintain, and secure APIs at nearly any scale, across any number of Regions. DoorDash replicated its API across three Regions in the United States to distribute the load more evenly, provide redundancy, and avoid downtime. The company also used the latency-based routing feature of Amazon Route 53, a reliable and cost-effective way to route end users to internet applications, so that users would automatically be routed according to their proximity to the appropriate Region.

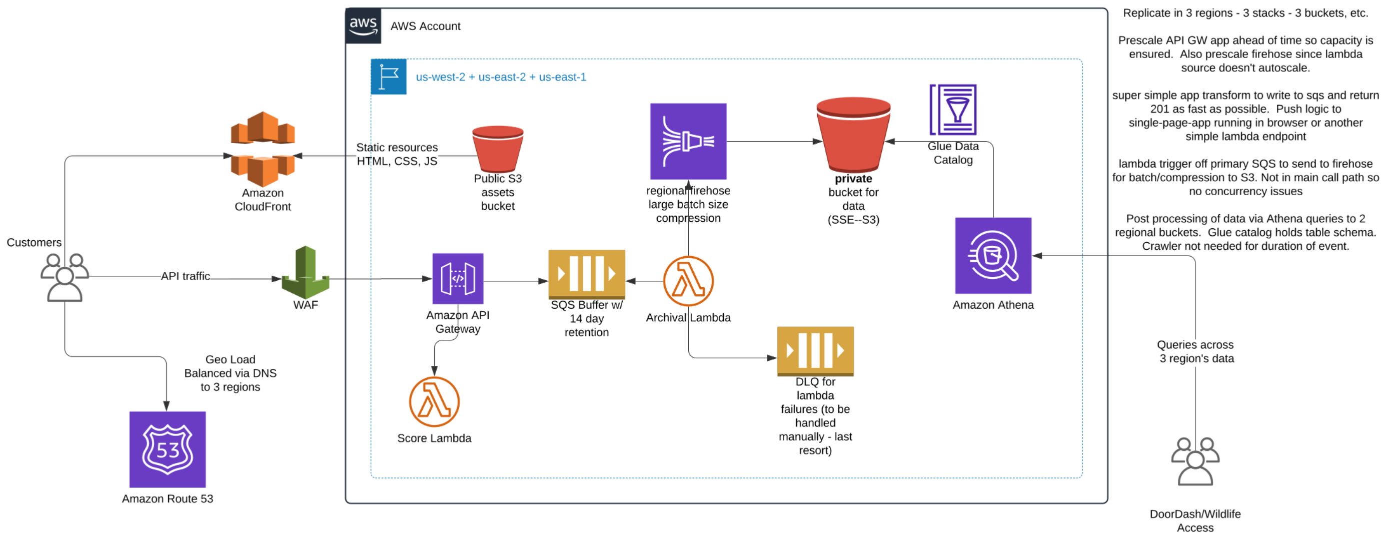
Wildlife and DoorDash performed stress tests until the night before the game, which revealed the importance of pre-scaling certain services to verify that the microsite was ready to deliver a fast, reliable user experience. “By pre-scaling the frequently used components in our load testing on recommendations from AWS, we felt more confident about the launch,” says Steve Hoffman, Software Engineer at DoorDash.

DoorDash connected Amazon API Gateway to Amazon Simple Queue Service (Amazon SQS), a fully managed message queue for microservices, distributed systems, and serverless applications, to decouple submission processing from the rest of the architecture. The company stored all data for the microsite using Amazon Simple Storage Service (Amazon S3), which provides object storage built to retrieve virtually any amount of data from anywhere.

To process the entries, DoorDash ran an AWS Lambda function to automatically retrieve data from Amazon SQS and send it to Amazon Kinesis Data Firehose, which reliably loads near real-time streams into data lakes, warehouses, and analytics services. “With the EDA, we had a lot of dashboards so that we could watch as the data moved through the environment to look for hot spots, both when we were testing and when the site was live,” says Hoffman. Meanwhile, to keep the frontend simple, the company took advantage of the distribution and scalability of Amazon CloudFront, which securely delivers content with low latency and high transfer speeds.

Architecture Diagram

DoorDash’s sweepstakes architecture on AWS



Click to enlarge for fullscreen viewing.

Outcome | Expanding the Use of EDAs for Future Projects

During the football championship broadcast, DoorDash handled more than 10 million entries on AWS. The EDA supported the huge spike in users. “We had no issues with handling the traffic during the event,” says Alexander. “Everything stayed up on AWS, and it was about as smooth as you could expect.” With this architecture, the site maintained its fast performance.

This build demonstrated how an EDA and serverless technology can unlock efficient scalability and reliability. The company plans to look for more opportunities to use this architecture in future projects. “Using an EDA was a great learning experience because it proves that the technology is ready for prime time,” says Hoffman. “You can get advantages that you don’t get when you’re running your own servers.”

About DoorDash

Based in San Francisco, DoorDash is a local commerce platform that operates in more than 30 countries with 550,000 merchants, 37 million active monthly consumers, and 7 million Dashers.

AWS Services Used

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

Learn more »

Amazon API Gateway

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale.

Learn more »

Amazon Route 53

Amazon Route 53 is a highly available and scalable Domain Name System (DNS) web service. Route 53 connects user requests to internet applications running on AWS or on-premises.

Learn more »

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

Learn more »

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Airbnb su AWS

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2018

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McDonald's Corporation è una catena statunitense di fast food e hamburger che serve 69 milioni di clienti ogni giorno. Grazie ad Amazon Web Services (AWS), si è trasformata in un'azienda digitale in grado di superare i propri

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Domino's

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2020

UNITED STATES

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Expedia Global Payments ha migrato un sistema legacy da Microsoft SQL Server ad Aurora PostgreSQL, scalando in modo economico per soddisfare il traffico e fornendo dati quasi in tempo reale a utenti e team interni.

2021

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